Greenhouse Climate Controller

Run the greenhouse controls for a plant that cannot complain, only wilt.

Win condition: Best plant-zone-control match supported by tour evidence.

THE SCIENCE YOU ARE USING

Controlled environments. A greenhouse lets growers set temperature, humidity, and light to match what a plant needs. Get it wrong and the plant wilts, scorches, or molds. Every setting is a tradeoff: more light can mean more heat and faster drying.

Evidence from the tour. On the greenhouse tour you gather clues about real plant needs and real zone settings. Back at the table you match plant cards to climate dials using that evidence, then defend the layout.

HOW TO BUILD AND RUN IT

01 **Gather tour evidence.** During the greenhouse visit, note real temperature, humidity, and light clues for different zones.

02 **Read the plant cards.** Each plant card lists needs and tolerances. Sort cards by what they require.

03 **Set the climate dials.** Match each plant to a zone and set temperature, humidity, and light on the dial board.

04 **Resolve tradeoffs.** When two plants want different things in one zone, choose the setting your evidence best supports.

05 **Defend the layout.** Explain each placement using a specific clue from the tour.

Safety: Stay with tour leader. Do not touch restricted plants or greenhouse controls. Allergy: Plant and pollen sensitivity note.

MATERIALS

Plant profile cards	set
Control dial boards	per team
Clipboards	6
Dry erase markers	per team
Tour clue sheets	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________

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The ONE variable we are testing: ______________________

Baseline result: ______________________

After redesign: ______________________

DEFEND YOUR DESIGN

What evidence made you change your design? ______________________

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If you ran it again, what is the ONE thing you would change? ______________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct setting choices	30
Use of tour evidence	25
Explanation	20
Layout efficiency	15
Teamwork	10
Total	**100**